

CyberDefense Pro 2.0

2.6.3 MITRE ATT&CK Framework

This lesson covers the following topics:

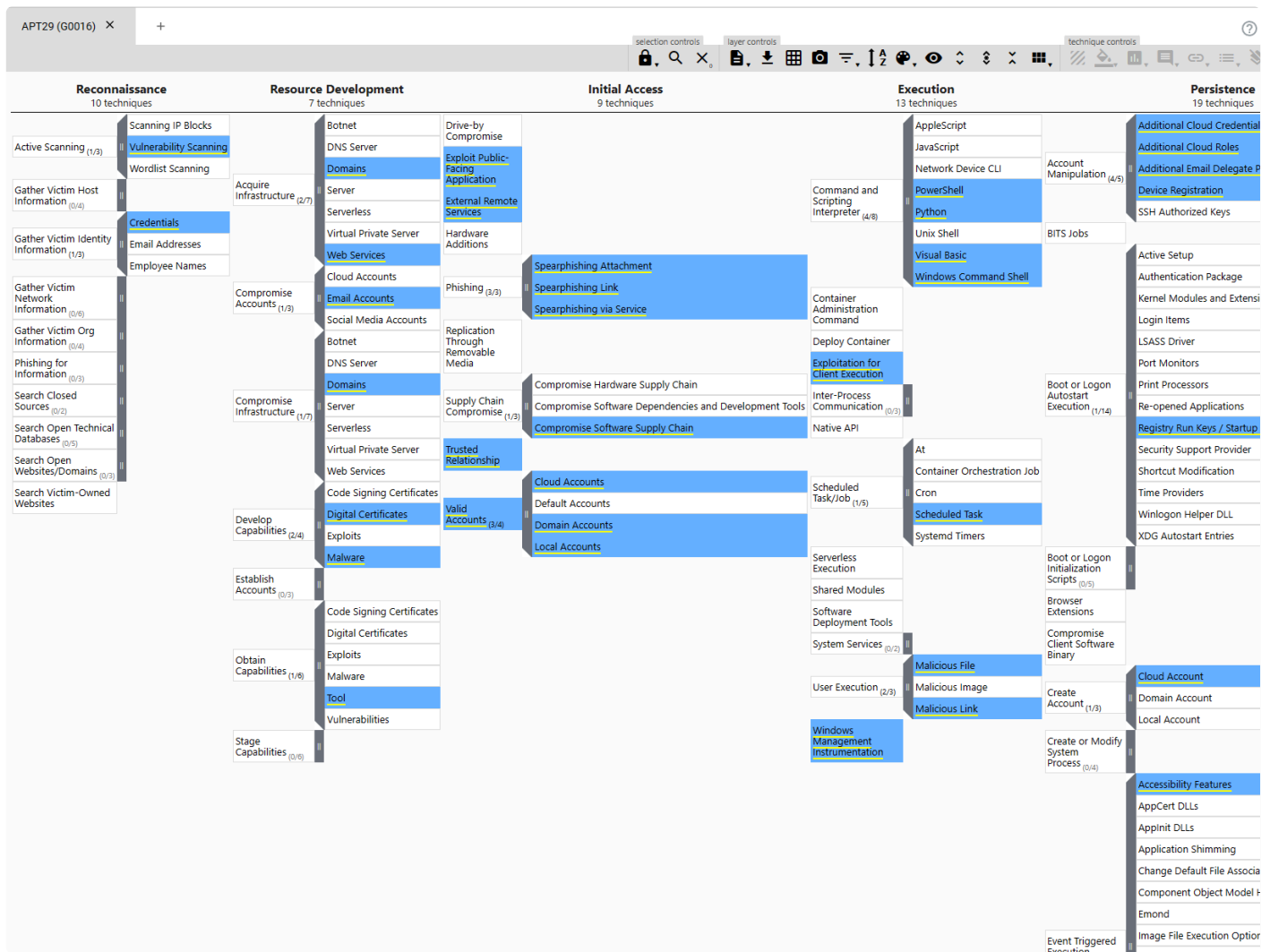
- Modified kill chain
- MITRE ATT&CK framework
- Developing a network threat model

Modified Kill Chain

As an early model, the kill chain has faced criticism for not accurately reflecting the chain of events in modern attack campaigns by focusing too much on perimeter security. Modified models, such as ones by AlienVault (now AT&T Cybersecurity) (cybersecurity.att.com/blogs/security-essentials/the-internal-cyber-kill-chain-model) and Sean Malone (blackhat.com/docs/us-16/materials/us-16-Malone-Using-An-Expanded-Cyber-Kill-Chain-Model-To-Increase-Attack-Resiliency.pdf), introduce iterative internal reconnaissance, lateral movement, privilege escalation, and data collection phases within Actions on Objectives. Once the attacker has achieved their initial objective without being detected, they may either maintain access or seek to remove any trace of their presence (anti-forensics).

The MITRE ATT&CK Framework

As an alternative to the lifecycle analysis described by a kill chain, the MITRE Corporation's Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK) matrices provide access to a database of known tactics, techniques, and procedures (TTPs). This freely available resource (attack.mitre.org) tags each technique with a unique ID and places it in one or more tactic categories, such as initial access, persistence, lateral movement, or command and control. The sequence in which attackers may deploy any tactic is not defined, so analysts must interpret each attack lifecycle from local evidence. The framework also allows analysts to compare the TTPs used by different threat groups.



▼ Description

The image is a screenshot from the MITRE ATT&CK Navigator tool, displaying the ATT&CK matrix specifically filtered for the APT29 threat group - commonly attributed to Russia's Foreign Intelligence Service. The matrix is organized into columns representing different attack tactics across the top, such as Reconnaissance, Resource Development, Initial Access, Execution, Persistence, and others. Within each column, specific techniques associated with each tactic are vertically listed.

Some techniques are highlighted in blue, marking those reportedly used by APT29. These include, but are not limited to, "Vulnerability Scanning," "Credentials," "Domains," "Spearphishing Attachment," "PowerShell," "Windows Command Shell," "Scheduled Task," and "Registry Run Keys / Startup Folder," among others. Each technique is identifiable by its name, and some techniques expand into additional sub-techniques (noted by downward arrows or indentation).

Above the matrix, there are navigation controls, search boxes, and visualization tools typical of web-based dashboards. The purpose of the matrix is to map out the sequence

and variety of adversary behaviors (TTPs) rather than a rigid, step-by-step order; this allows analysts to see all known techniques used by APT29 and supports comparison across multiple threat groups.

The uploaded screenshot serves as a visual summary of which attack procedures have been attributed to APT29 within the official MITRE ATT&CK database, which is accessible for further exploration at attack.mitre.org.

Developing a Network Threat Model

Threat actors represent a person, group, or organization responsible for malicious activities. They are often motivated by financial gain, political gain, or simply a desire to cause harm. This lesson reviews different threat actor types, their tactics, and methods to identify and defend against them. Organizations and individuals can be more aware and better prepared to fend off malicious attacks by exploring threat actor concepts.

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